

EARTHQUAKE-PROOF BUILDING CHALLENGE

Grade 4 STEM Exploration Guide: Let's Learn How to Build Strong Towers!



What Happens During an Earthquake?

Earthquakes make the ground shake with waves called **seismic waves**! These waves travel through the Earth like ripples in a pond. When they reach the surface, they make everything move and shake. That's why buildings need to be strong enough to handle these shaking waves!



3 Key Principles of Safe Buildings

- **Triangles:** Triangle shapes keep buildings strong, rigid, and steady.
- **Cross-Bracing:** X-shaped supports stop buildings from wobbling back and forth.
- **Wide Base:** A wide, heavy base helps towers stand tall and resist shaking.



Local Wisdom: Amazing Indian Earthquake-Proof Buildings

- **Kath-Kuni Architecture:** Traditional wooden-stone buildings from Himachal Pradesh use flexible joints that sway gently instead of breaking during earthquakes!
- **Bhuj Reconstruction:** After the 2001 earthquake in Bhuj, Gujarat, engineers rebuilt homes using base isolation and cross-bracing to create super safe buildings.

STEM Insight: Allowing a building to *flex and sway* slightly absorbs seismic energy better than making a rigid building that might snap!



Let's Build Our Own Tower! (STEM Activity)

Get ready to become a structural engineer! Work with a partner to design and construct your own earthquake-proof tower using toothpicks/straws and marshmallows or clay.

1. **Step 1:** Build a wide, solid base first so your tower won't tip over easily.
2. **Step 2:** Add vertical height using triangles and X-shaped cross-bracing for strong support.
3. **Step 3:** Test your tower on a shaking tray or jelly plate to see if it stands tall!
4. **Step 4 (Show Off!):** Share your design with classmates and discuss what made your tower sturdy.

Practice Questions: Earthquake-Proof Buildings

Test your STEM knowledge! Answer the 12 questions below.

Student Name: _____

Date: _____

1. What are the waves that cause the ground to shake during an earthquake called?

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2. How do seismic waves travel through the Earth?

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3. Which geometric shape is known for making building structures strong and steady?

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4. What type of supports prevent a building from wobbling from side to side?

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5. Why is having a wide base important for a tall tower during an earthquake?

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6. What traditional architectural style from Himachal Pradesh uses wooden-stone joints to sway safely?

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7. In which Indian region were homes rebuilt using base isolation after the 2001 earthquake?

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8. True or False: A building should be completely stiff with zero flexibility to survive an earthquake.

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9. What step should you complete first when constructing your model earthquake tower?

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10. Name two materials you can use at school or home to build a test model tower.

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11. Why is cross-bracing often shaped like the letter "X"?

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12. Imagine your test tower falls over during a shake test. What structural change would you make to fix it?